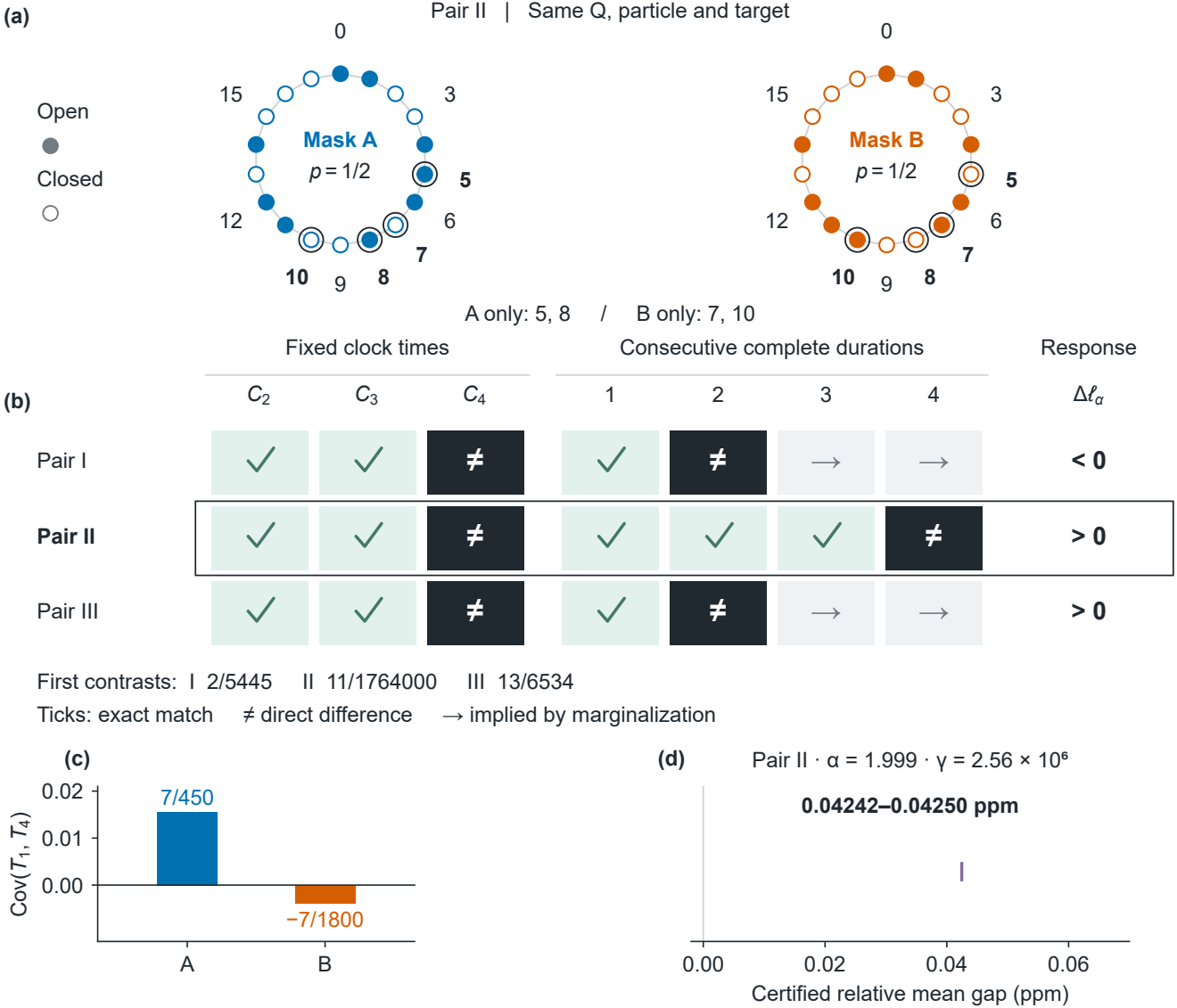

(a)
Pair II | Same Q, particle and target
Open
Closed
Mask A
p = 1/2
Mask B
p = 1/2
0
3
5
6
7
8
9
10
12
15
A only: 5, 8 / B only: 7, 10
(b)
Fixed clock times
Consecutive complete durations
Response
C2
C3
C4
1
2
3
4
Δℓα
Pair I
Pair II
Pair III
< 0
> 0
> 0
First contrasts: I 2/5445 II 11/1764000 III 13/6534
Ticks: exact match ≠ direct difference → implied by marginalization
(c)
Cov(T1, T4)
0.02
0.01
0.00
7/450
−7/1800
A
B
(d)
Pair II · α = 1.999 · γ = 2.56 × 10⁶
0.04242–0.04250 ppm
0.00
0.02
0.04
0.06
Certified relative mean gap (ppm)